THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME NOVEMBER 2025

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1. (a) $= \frac{7}{8} \div \left(\frac{3}{2} \times \frac{14}{9}\right)$ .....01 mark.

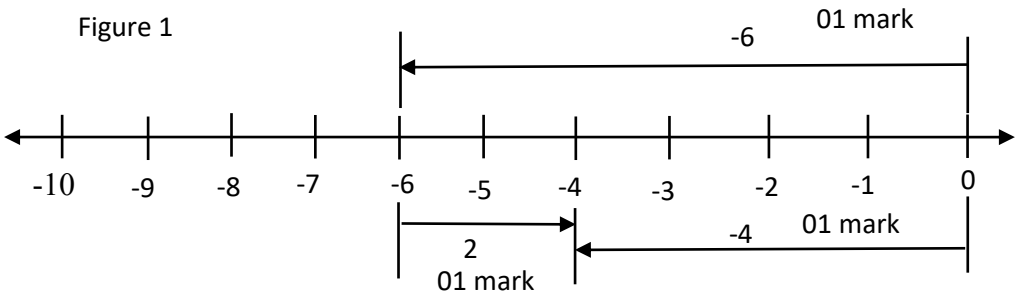
$= \frac{7}{8} \div \frac{7}{3}$ .....01 mark

$= \frac{7}{8} \times \frac{3}{7}$ .....01 mark.

$= \frac{3}{8}$ .....02 marks

(b) $-6 + 2 = -4$

Figure 1



$\therefore (-6) + 2 = (-4)$ .....02 marks

2. (a) i. $2.555555 \ldots = 2.\dot{5}$ .....$00\frac{1}{2}$ mark.

ii. $0.212121212 \ldots = 0.\overline{21}$ .....$00\frac{1}{2}$ mark.

iii. $1.15151515 = 1.\overline{15}$ .....$00\frac{1}{2}$ mark.

iv. $5.0000300003 \ldots = 5.\overline{00003}$ .....$00\frac{1}{2}$ mark.

(b) Let m *be* $0.8\dot{3}$

$m = 2.\dot{5}$ .....equation 1

$10m = 2.\dot{5} \times 10 = 25.\dot{5}$ .....equation 2

$$\begin{array}{l|l} - & 10m = 25.\dot{5} \\ & m = \quad 2.\dot{5} \end{array}$$

$9m = 23$ .....`01 mark.

$\frac{9m}{9} = \frac{23}{9}$

m= $\frac{23}{9}$ .....01 mark.

Let y *be* $0.\overline{21}$

$y = .0.\overline{21}$ .....equation 1

$100m = \times 0.\overline{21} \times 100 = 21.\overline{21}$ .....equation 2

$$-\left|\begin{matrix}100y = 21.\overline{21}\\ y = \quad 0.\overline{21}\end{matrix}\right.$$

$99y = y = 21$……`01 mark.

$\frac{99y}{99} = \frac{21}{99}$

$= \frac{21}{99} = \frac{21\div 3}{99\div 3} = \frac{7}{33}$……01 mark.

$2.\dot{5} \times 0.\overline{21} = \frac{23}{9} \times \frac{7}{33} = \frac{161}{297}$……01 mark.

(c) i. $34.996 \approx 35.00$……01 mark.

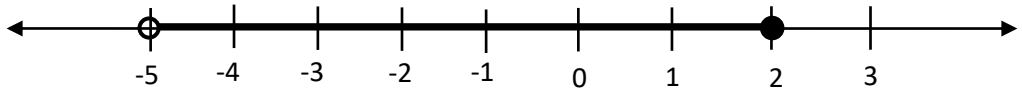
ii. $35.0483 \approx 35.0$……01 mark.

3.(a) 20 192 + 18 800 + 23 468 +25 477 +18 991+260 060+ 22 645$= 389\ 633$ ….$02\frac{1}{2}$ marks.

$389\ 633 \approx 390\ 000$……$02\frac{1}{2}$ mark.

(b) Figure 2

$-5 < y \leq 2$



4. (a) $\frac{1}{2}\left|x + \frac{y}{2} = 5\right.$

$1\left|\frac{x}{2} - \frac{y}{3} = \frac{1}{6}\right.$

$\left|\begin{matrix}\frac{x}{2} + \frac{y}{4} = \frac{5}{2}\\ \frac{x}{2} - \frac{y}{4} = \frac{1}{6}\end{matrix}\right.$

$-\left|\begin{matrix}\frac{x}{2} + \frac{y}{4} = \frac{5}{2}\\ \frac{x}{2} - \frac{y}{3} = \frac{5}{2}\end{matrix}\right.$ ……01 mark.

$\frac{y}{4} + \frac{y}{3} = \frac{5}{2} - \frac{1}{6}$

$\frac{3y+4y}{12} = \frac{15-1}{6}$

$\frac{7y}{12} = \frac{14}{6}$……01 mark.

$y = \frac{14\times 12}{7\times 6}$

$x = 4$……01 mark.

Substitute $4\ for\ y\ in\ x + \frac{y}{4} = 5$

$x + \frac{4}{2} = 5$

$x + 2 = 5$

$x = 5 - 2$ ……01 mark.

$x = 3$……01 mark.

(b) Total parts $= 5 + 3 + 2 = 10$parts……01 mark.

Difference in shares $= \left(\frac{5-2}{10}\right) \times 60\,000$ shillings……01 mark.

$= \frac{3}{10} \times 60\,000$ shillings……01 mark.

$= 18\,000$ shillings……02 marks.

5. (a) $m = \frac{y_2 - y_1}{x_2 - x_1}$……$00\frac{1}{2}$ mark.

$= \frac{4-2}{4+3} = \frac{2}{7}$……01 mark.

$\frac{2}{7} = \frac{y-2}{x+3}$……$00\frac{1}{2}$ mark..

$2x + 6 = 7y - 14$

$2x + 6 + 14 = 7y - 14 + 14$

$2x + 20 = 7y$

$\frac{7y}{7} = \frac{2x+20}{7}$ or $2x - 7y + 20 = 0$ ……01 mark.

$y - intercept = c$ $from$ $y = mx + c$ ……01 mark.

Therefore $y - intercept = c = \frac{20}{7} or \left(0, \frac{20}{7}\right)$……01 mark.

(b) $\pm(x - 3) \leq 5$……01 mark.

$x - 3 \leq 5$

$x - 3 + 3 \leq 5 + 3$

$x \leq 8$ ……01 mark.

$or - (x - 3) \leq 5$

$-x + 3 \leq 5$

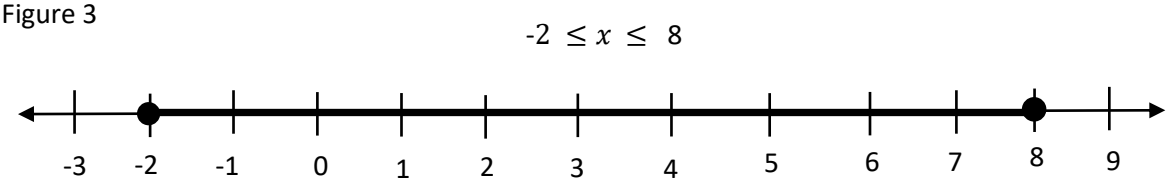
$-x + 3 - 3 \leq 5 - 3$

$-x \leq 2$

$\frac{x}{-1} \geq \frac{2}{-1}$

$x \geq -2$……01 mark.

Figure 3



6.(a) $1m = 100\ cm$

$5m = y$

$y = \frac{5\ m \times 100\ cm}{1m}$

$y = 500\ cm$……01 mark.

$200\ cm : 5m = 200\ cm : 500\ cm$

$= \frac{200\ cm}{500\ cm} = \frac{2}{5}$ ……02 marks.

$200\ cm : 5\ m = 2 : 5$……02 marks.

(b)

x	-3	-2	-1	0	1	2	3

y	4	3	2	1	0	-1	-2

.....05 marks.

7. (a) i. $\frac{1}{q}(qz + 4qz) = \frac{qz}{q} + \frac{4qz}{q}$ .....00$\frac{1}{2}$ mark.

$= z + 4z$ .....01 mark.

$= 5z$ .....01 mark.

ii. $\frac{2}{x} + 3 - 3 = 4 - 3$ .....00$\frac{1}{2}$ mark.

$\frac{2}{x} = 1$ .....01 mark.

$2 = x$ .....01 mark.

(b) Let the numbers be $x \; and \; y$

$x + y = 86$ .....equation 1.....00$\frac{1}{2}$ mark.

$x - y = 14$.....equation 2.....00$\frac{1}{2}$ mark.

$+ \begin{vmatrix} x + y = 86 \\ x - y = 14 \end{vmatrix}$

$2x = 100$ .....01 mark.

$\frac{2x}{2} = \frac{100}{2}$

$x = 50$ .....01 mark.

Substitute $50 \; for \; x \; in \; x - y = 14$

$50 - y = 14$ .....01 mark.

$y = 50 - 14 = 36$.....01 mark.

Therefore, the numbers are 50 and 36

8. (a)Using $x - intercept \; and \; y - intercept$

$2x + y = 5$

x	y
0	5
$2\frac{1}{2}$	0

$x + y = 3$

x	y
0	3
3	0

.....02 marks.

Alternative: $y = -2x + 5$

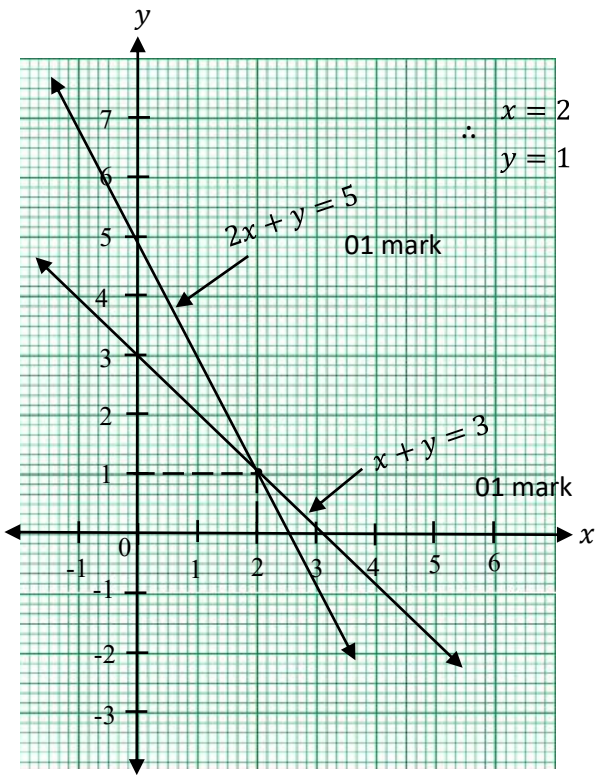
x	-2	-1	0	1	2
y	9	7	5	3	1

$y = -x + 3$

x	-2	-1	0	1	2
y	5	4	3	2	1

.....02 mark.

Figure 5



01 mark

(b) Multiples of 2 are 2, 4, 6, 8, 10, 12, 14, 16,… ……..01 mark.

Multiples 0f 7 are 7, 14, 21,……..01 mark.

Common multiples is 14……..01 mark.

Second multiple $= 14 \times 2 = 28$ ……..01 mark.

Third multiple $=14\times 3 = 42$……..01 mark.

Therefore, the common multiples of 2and 7 between0-50 are 14, 28, and 42

9. (a) $206 \times 206 = 42436$……$02\frac{1}{2}$ marks.

Didit2 is at thousands place value or position……$02\frac{1}{2}$ marks.

(b)**Arithmetic**: is the branch of Mathematics that deals with properties and manipulation of numbers by using basic Mathematical operations such as addition, subtraction, multiplication and division……02 marks.

Algebra: is the branch of Mathematics in which arithmetic operations are applied to symbols rather than specific numbers……$1\frac{1}{2}$ marks

Geometry: is the branch of Mathematics which deals with the study of the sizes, shapes, positions, angles and dimensions of different physical objects……$1\frac{1}{2}$ marks.

10. (a) $\frac{5}{100} \times 300\,000 = 15\,000$ ……$1\frac{1}{2}$ marks.

$\frac{5}{100} \times 600\,000 = 210\,000$……$1\frac{1}{2}$ marks.

$210\,000 - 15\,000 = 195\,000$……02 marks.

(b) $(-6\,, n) \rightarrow (x\,, y)$

$3(-6) - 4(n) - 6 = 0$……01 mark.

$-18 - 4n - 6 = 0$

$-18-6-4n=0$

$-24-4n=0$……01 mark.

$-4n=24$……01 mark.

$\frac{-4n}{-4}=\frac{24}{-4}$

$n=-6$……01 mark.